MyungJin Ko (Ray)

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Summary

Full-stack engineer with 7 years of experience, including 5 as co-founder and founding engineer of a B2B SaaS startup. Designed, built, and operated a multi-tenant SaaS from zero to 10+ business customers — absorbing event-day peaks of ~25K concurrent users through pre-rendered, CDN-cached read paths, PostgreSQL tuning, and pre-event load testing — while delivering 20–30 event platforms a year for national institutions. Primary stack is TypeScript; I've shipped production systems in Rust, Java, and C# where the problem called for it, and I operate what I ship, end to end.

Skills

- **Fluent**: TypeScript, JavaScript, React, Next.js, Node.js, NestJS, PostgreSQL
- **Worked with**: Rust (Axum), Java / Spring, C# / .NET (Revit), React Native, Tauri, Python, Express, MySQL
- **Data & search**: PostgreSQL, Supabase, Elasticsearch
- **Infrastructure**: AWS (ECS, EC2, RDS, Lambda), Fly.io, Vercel, Cloudflare, Docker, NHN Cloud, CI/CD, Sentry, PostHog
- **Areas**: multi-tenant SaaS, real-time systems, REST API design, performance optimization, 0-to-1 product

Experience

Plancy — Co-founder & Founding Engineer

Seoul, South Korea · May 2021 – Jun 2026

Plancy ran two lines of business for the MICE (events & exhibitions) industry — its own multi-tenant SaaS, and a web agency building event websites and registration platforms for event organizers (roughly KRW 1B / ~$750K in cumulative revenue across both). I led engineering and infrastructure for both.

MICE All-in-One — B2B multi-tenant SaaS (mice.ai.kr) · Lead Engineer · Feb 2024 – Jun 2026 Led engineering in a 3-person team (a PM/designer and a junior engineer).

- **0→1 ownership**: built the product from PoC to production for 10+ agency customers. Modeled each event as the tenant unit so infrastructure cost scaled with projects rather than end users, and aligned per-project pricing to that cost driver (KRW 200–300M / ~$150–220K ARR) — keeping the service profitable as customers ran more events.
- **Spiky, event-driven load**: absorbed event-day peaks of ~25K concurrent users against a ~1K baseline DAU across 3–4 large government-backed events per year, without over-provisioning — pre-rendered and CDN-cached the read paths to keep load off the origin, tuned PostgreSQL indexes and connection pooling, and validated capacity with pre-event load tests. Zero downtime across those events.
- **Performance & reliability ownership**: diagnosed the platform's worst bottleneck — bulk on-demand document generation timing out on serverless — and moved that hot path to a dedicated Rust (Axum) service via a strangler-fig migration, chosen for the CPU-bound rendering workload and its memory footprint on small instances. Eliminated failures on large batch jobs and instrumented monitoring to detect and re-evaluate bottlenecks as they surface.

- **Identity & tenancy**: consolidated three separate login flows into one identity model with per-event tenant isolation (PostgreSQL RLS), and shipped self-service domain provisioning — organizers claim an auto-deduplicated subdomain or connect a purchased domain, configured automatically via the Cloudflare API.
- **Platform surface**: opened the product to integrations with a public REST API and scoped API keys (a developer center), and built a JSON-Schema–driven no-code form builder for custom registration and booth-application forms.
- **Event-operations toolkit**: real-time business matching (meeting requests, live chat, scheduling), QR-based on-site check-in with registration-to-on-site funnel analytics (exportable), and scheduled/recurring invoicing and email with delivery and open tracking (cron + Resend).
- **Quality & release safety**: automated tests on critical flows plus preview deployments and rollback procedures kept releases safe with a 2-engineer team; wired Sentry and PostHog observability into an internal admin console giving leadership a single view of platform health.
- **AI-native workflow**: established an AI-agent–centric development workflow (Cursor, Claude Code) that let a small team sustain output across two product lines, and prototyped a RAG-based proposal assistant to validate AI features in the product.

Stack: TypeScript, Next.js, React, Node.js, Rust (Axum), Supabase (PostgreSQL, RLS, Realtime), Cloudflare, Docker, Fly.io, Vercel, TipTap/YJS, Tauri, Sentry, PostHog, GitHub Actions

Event Web & Registration Platforms — national institutions & large events · Technical Lead · Oct 2021 – Oct 2025

- Worked directly with stakeholders at national institutions (e.g., National Hangeul Museum, Sculpture City Seoul) to **scope requirements and deliver end to end** — websites and real-time registration platforms, from planning through production operations.
- Delivered 20–30 events per year (each ~2,000 DAU at peak registration) over 3–4 years; templatized the recurring site and registration flows to cut build time for each new event.
- Owned infrastructure and deployment, plus the team's internal dev environment, tooling, and workflow.

Stack: Next.js, TypeScript, React, Node.js, Supabase/PostgreSQL, Vercel, Cloudflare, Docker

Client Engineering (enterprise SI & prototyping) · Software Engineer / Tech Lead · Oct 2021 – Oct 2025

- Delivered 3–6 month projects for enterprise clients, adapting stack and architecture per project — from frontend-only apps to serverless and full-stack systems.
- **Solar-design review platform**: built an Autodesk Revit add-in (C# / .NET) to extract BIM data, designed a custom file format (.pim, a ZIP of structured JSON), parsed it via Next.js into Supabase, and surfaced reports, charts, and national-standard compliance checks — an end-to-end desktop-CAD-to-web pipeline.
- **Public-sector lookup system**: Java / Spring on a CSAP-certified cloud (NHN Cloud), with PII encryption and access-control design.

Stack: C#/.NET, Autodesk Revit API, Next.js, TypeScript, Java, Spring Boot, NHN Cloud, Supabase

The Scent Lab — Co-founder & Founding Engineer

Seoul, South Korea · Sep 2020 – May 2021

My first startup. Led product and frontend, and built the server alongside a junior backend engineer I mentored.

- Built and launched a personalized fragrance recommendation app (React Native / Expo) on iOS and Android.
- Designed the recommendation engine over a ~70K-item dataset structured by scent notes — Elasticsearch weighted scoring combined with survey-based personalization.
- Designed and deployed the NestJS backend as containers on AWS ECS.

- Co-filed a patent for the recommendation process; secured ~KRW 50M in Korean government startup funding.

Stack: React Native, Expo, TypeScript, NestJS, Elasticsearch, PostgreSQL, AWS (ECS, EC2, RDS, Lambda)

Supaja — Frontend Lead (contract) · Apr 2020 – Sep 2020

- Designed and built the service's entire frontend solo; onboarded one developer new to React and left structure and docs so the service ran without me.

Weplanet — Web Developer Intern · Mar 2019 – Aug 2019

- Web prototyping across client projects — React frontends with Node (Express) and MySQL backends.

Education

Sahmyook University — B.S. in Computer Science (Software major), 2021

Languages

Korean (native) · English (professional reading/writing; conversational speaking, improving)

Additional

- Co-filed patent — personalized recommendation process (The Scent Lab)
- AUSG / AWSKRUG (AWS Korea user group) community member (2019)